Midterm

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Homework Set Up

Reading in packages:

```
library(tidyverse) # load in all packages
library(janitor)
library(flextable) # new package to create a table
library(rstatix) # new package for effect size
```

Reading in data:

```
tussocks <- read_csv("tussock_height.csv") # store the data and save object as "tussocks"
temp <- read_csv("temp_2024.csv") # store the data and save object as "temp"
```

Problem 1. Understanding and critiquing written communication

a. Hypotheses

Null hypothesis (H_0): There is no significant difference in mean gonad weight of purple sea urchins between heated and ambient temperature treatments.

Alternative hypothesis (H_A): There is a significant difference in mean gonad weight of purple sea urchins between heated and ambient temperature treatments.

b. Test Type

For the researchers to run a student's two-sample t-test, two independent populations would need to be compared (one group of urchins in heated conditions, one in ambient temperature conditions), and have equal variances. Additionally, the variable of interest, weight, must be continuous, the weights must be randomly sampled, and have an approximately normal distribution.

c. Test Summary components

- t: This refers to the t distribution which is the sampling distribution of the test statistic comparing two population means in units of standard error. It is used because the population standard deviations of urchin gonad weight (in grams) are unknown.
- 53: This is the degrees of freedom for the t-distribution which is calculated by subtracting 2 from the total sample size across both groups ($n_{heated} + n_{ambient} - 2$). 53 is number of independent values that the statistical analysis can estimate, and determines the shape of the t-distribution used to assess statistical significance.
- -13.51: This is the calculated value of the test statistic, which indicates the difference in sample mean gonad weight between the two temperature conditions measured by units of standard error. The large absolute value represents a large difference between heated and ambient groups.
- $p < 0.001$: This is indicating that the probability of observing such extreme differences in mean gonad weights by random chance is less than 0.1% ($p < 0.001$). This leads to the assumption that the null hypothesis is rejected and that the purple sea urchins in each group come from different populations.

d. Missing Information

In this research report, the sample sizes for each treatment group, n, is missing. Knowing the sample size for heated and ambient groups is important for assessing whether the study had adequate statistical power to detect true differences between temperature conditions and whether the study design was balanced. Unequal sample sizes could affect the reliability of conclusions about climate impacts on uni production.

Another component that could be included in this report is the significance level (α). This component is relevant for addressing the hypotheses because it clarifies the predetermined threshold for deciding that the difference in mean gonad weight between each group is statistically significant. The significance level helps avoid bias and Type I error in hypothesis testing which is relevant in determining the biological significance in uni production.

Problem 2. Interpretation and communication

a. Hypotheses

The null hypothesis (H_0): There is no significant difference in the mean crop value (measured in 1000s of US dollars) across 28 crops in Kern County, CA between 2022

and 2023.
The alternative hypothesis (H_A): There is a significant difference in the mean crop value(measured in 1000s of US dollars) across 28 crops in Kern County, CA between 2022 and 2023.

b. Statistical summary

Within the supplementary data table (Table S3) in the paper and the test output given, two pieces of information that are the same include the t-value (t = -1.5) and the p-value (p = 0.14 or 0.1399).

Two pieces of information that the test output provides, but the researchers do not, are the 95% confidence interval (-51,595,337 to 7,664,551) and the mean difference (-21,965,393).

The confidence interval, -51,595,337 to 7,664,551, is the range of values in which the true mean difference in total crop value between 2022 and 2023 will exist 95% of the time. Because this interval includes zero, it indicates that the true mean difference could be zero. This indicates that we cannot be confident that crop values actually changed between the two years. The test output also provides the degrees of freedom (27). The degrees of freedom reflect the 28 total paired crop observations minus 1. This determines the shape of the t-distribution used to calculate the p-value.

c. Writing a new statement

The mean crop value (measured in 1000s of US dollars) across 28 crops in Kern County, CA between 2022(n=28) and 2023(n=28) was measured, and a significant difference was not found(Paired t-test, $t(27) = -1.52$, $p = 0.14$, $\alpha = 0.05$).

d. Statistical implications

The results of this study suggest that there is no significant difference in the mean crop value between 2022(extreme drought) and 2023 (extreme rainfall). 28 crops across Kern County, CA were measured and ultimately it was found that crop producers were, on average, not substantially negatively impacted by “hydroclimate volatility/ precipitation whiplash”. This suggests that the crop producers in Kern-County were able to adapt from the abrupt shift from a dry-to-wet year by employing land management strategies. It is possible that the benefits of extreme precipitation in water-scarce environments may be leveraged to decrease the economic costs during drought periods. I would suggest that blueberry farmers in Kern County use similar mitigation methods such as fallowing fields, using deficit irrigation, and crop-switching.

Problem 3. Reproducing an analysis

a. Variables

- The response variable: Average tussock diameter measured in cm, which is a quantitative, continuous variable.
- The predictor variable is the site type (burned or unburned) within the 2007 Anatuvuk River Fire scar, which is a binary categorical variable.
- The response variable measurement: The researchers measured the average tussock diameter along four 100 meter transects that were located in each cardinal direction from the meteorological tower. The tussocks that intercepted the transect tape were measured in two directions (N-S and E-W) with a caliper (Methods section, paragraph 5 (Microtopography)).

b. Hypotheses

The null hypothesis (H_0): There is no significant difference in the mean tussock mound diameters (cm) in burned and unburned sites.
The alternative Hypothesis (H_A): There is a significant difference in the mean tussock mound diameters (cm) in burned and unburned sites.

c. Cleaning and organizing data

```
tussocks_clean <- tussocks |> # start with tussocks data, create new object with clean data
  clean_names() |> # clean column names (lower case with underscores)
  select(site, average_diameter) |> # select only the site and average_diameter columns
  mutate(site = case_when(
    site == "Sev" ~ "Burned", # change "Sev" site name to "Burned"
    site == "Unb" ~ "Unburned")) # change "Unb" site name to "Unburned"

tussocks_clean |>
  slice_sample(n = 10) # displays 10 random rows from the "tussocks_clean" data-frame

# A tibble: 10 × 2
   site     average_diameter
   <chr>               <dbl>
 1 Burned               27.1
 2 Unburned             21.1
 3 Burned               22.4
 4 Unburned             17.9
 5 Burned               24.5
```

```
 6 Unburned          26.4
 7 Unburned          17.1
 8 Unburned          28.1
 9 Unburned          22.5
10 Burned           24.6
```

d. Summarizing and table displays

```
tussocks_summary <- tussocks_clean |> # create an object called tussocks_summary using the tussocks_clean data frame
  group_by(site) |>  # group by site
  summarize(
    mean = round(mean(average_diameter, na.rm = TRUE), digits = 1), # calculates the mean and rounds the value to 1 decimal place
    se = round(sd(average_diameter, na.rm = TRUE) / sqrt(n()), digits = 1) # calculates standard error and rounds the value to 1 decimal place
)

# create polished table using flextable
tussocks_summary |> # pipe the tussocks_summary data frame into the table
  flextable() |> # flextable() converts the data frame into a formatted table object
  # renames the column headers to be more readable
  set_header_labels( site = "Site",
                     mean = "Mean",
                     se = "Standard Error") |>
  autofit() # automatically adjusts column widths to fit the content properly
```

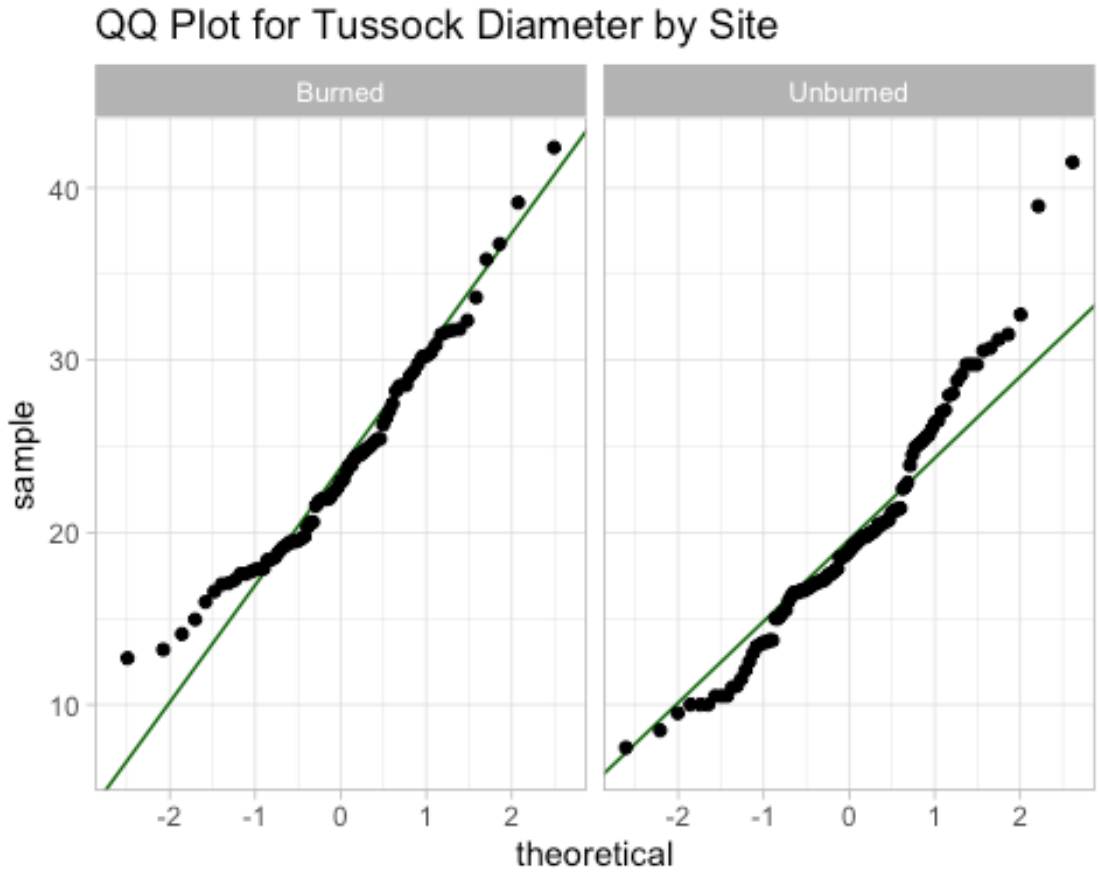
Site	Mean	Standard Error
Burned	23.8	0.7
Unburned	19.7	0.6

e. Check your assumptions

Check 1: Visual tool - Normally distributed variable

```
ggplot(data = tussocks_clean,
       aes(sample = average_diameter)) + # specifies the variable to check for normality
  geom_qq_line(color = "darkgreen") + # making reference line green so its easier to see
  geom_qq() + # geom_qq() plots the QQ points comparing data to theoretical normal
```

```
distribution
  facet_wrap(~site) + # creates separate plots for each site (Burned and Unburned)
  labs(title = "QQ Plot for Tussock Diameter by Site") + # adds title to the plot
  theme_light() # applies a minimal theme to the plot
```



Visual check: The points on the QQ plot for both burned and unburned sites approximately follow the reference line with only minor deviations at the tails. The reference line represents the theoretical distribution, so tussock diameters appear normally distributed.

Check 2: Equal Variance Test

```
var.test( # perform an F-test to compare variances in tussock diameters
  average_diameter ~ site, # test whether variances of tussock diameters differ by site
  data = tussocks_clean) # use the cleaned tussock data-set
```

```
  F test to compare two variances

data:  average_diameter by site
F = 0.92558, num df = 78, denom df = 110, p-value = 0.7221
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.6169823 1.4090774
sample estimates:
ratio of variances
       0.9255843
```

Using a F-test of equal variance, I determined that group variances are equal (F-test, F(78, 110) = 0.93, p = 0.72).

f. Do a t-test

```
t.test( # perform a two-sample t-test to compare mean tussock diameter
  average_diameter ~ site, # test whether mean tussock diameters differ by site
  var.equal = TRUE, # assume equal variances based on the variance test
  data = tussocks_clean # use the cleaned data frame again
)


  Two Sample t-test

data:  average_diameter by site
t = 4.4663, df = 188, p-value = 1.371e-05
alternative hypothesis: true difference in means between group Burned and group Unburned is
not equal to 0
95 percent confidence interval:
 2.298075 5.934027
sample estimates:
 mean in group Burned mean in group Unburned
             23.78632             19.67027

 # cohens_d() calculates Cohen's d effect size
 cohens_d(
  # calculate effect size for average diameter between sites
  data = tussocks_clean, # use the cleaned tussock data
  average_diameter ~ site,
  var.equal = TRUE) # matches t-test assumption of equal variances
```

```
# A tibble: 1 × 7
  .y.           group1 group2   effsize    n1    n2 magnitude
* <chr>         <chr>  <chr>      <dbl> <int> <int> <ord>
1 average_diameter Burned Unburned   0.657    79   111 moderate
```

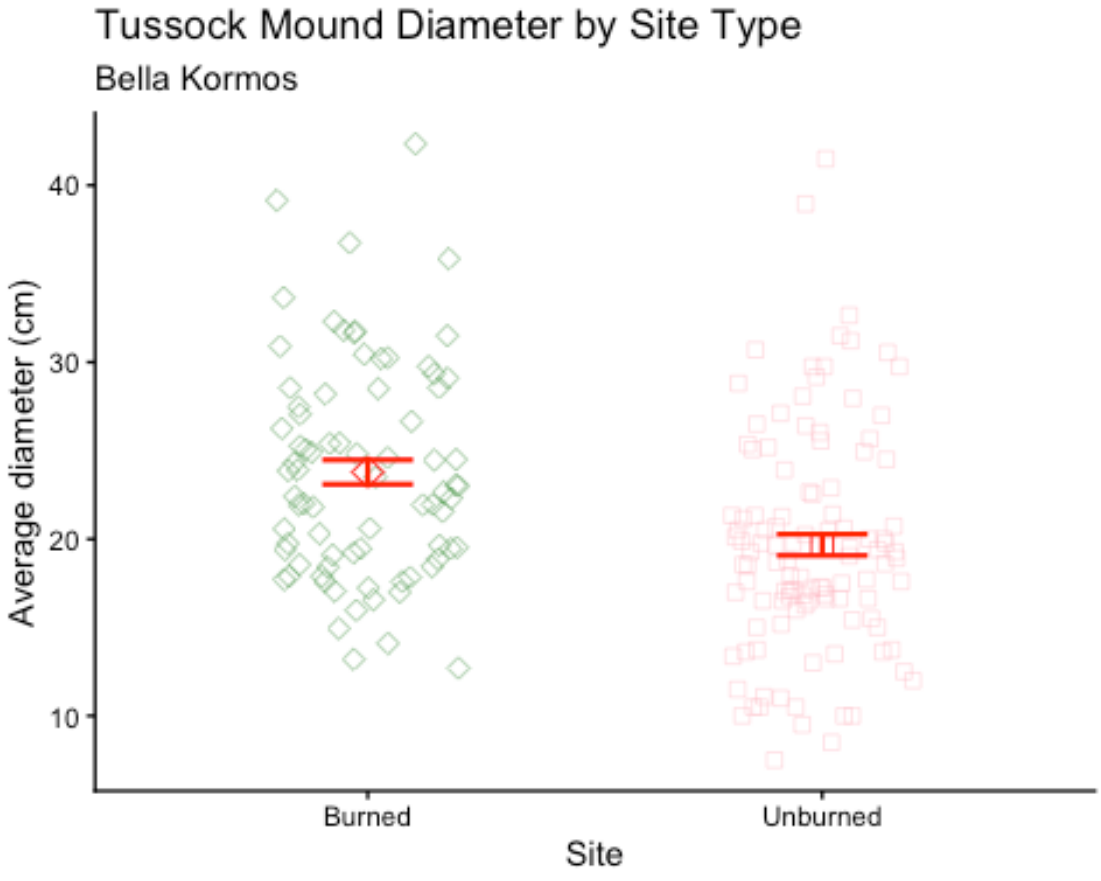
g. Written Communication

We found a significant difference in tussock mound diameters between unburned and burned sites (Student's t-test: $t(188) = 4.5$, $p < 0.001$, $\alpha = 0.05$) with a moderate effect size (Cohen's $d = 0.66$). On average, the mean tussock diameters were larger by about 4.1 cm in burned sites (mean = 23.8 cm, SE = 0.7, n = 79) in comparison to unburned sites (mean = 19.7 cm, SE = 0.6, n = 111).

h. Making a new figure

```
# create plot with x-axis as site, y-axis as average diameter
ggplot(data = tussocks_clean,
     aes(x = site,
        y = average_diameter,
        color = site,
        shape = site)) +
 # adds individual data points
 geom_jitter( width = 0.2, # jitters points horizontally
           height = 0, # prevents vertical jitter
           alpha = 0.4, # make transparent
           size = 2) + # edit site of points
 stat_summary(geom = "point", # adds mean as a point
           fun = mean, # calculates mean for each site
           color = "red", # makes mean point "red"
           size = 3) + # makes mean point larger than data points
 stat_summary(fun.data = mean_se, # calculates standard error and mean
           geom = "errorbar", # displays error bars
           width = 0.2, # sets width of error bar caps
           linewidth = 0.8, # sets thickness of error bar lines
           color = "red") + # colors error bars red
# adds plot title, author name as a subtitle, and labels axes
labs(title = "Tussock Mound Diameter by Site Type",
   subtitle = "Bella Kormos",
   x = "Site",
   y = "Average diameter (cm)") +
 # use a theme that creates white background without gridlines
```

```
theme_classic() +
# removes legend
theme(legend.position = "none") +
# sets custom colors for each site
scale_color_manual(values = c("Burned" = "forestgreen", "Unburned" = "lightpink")) +
# sets custom shapes (5 = open diamond, 0 = square)
scale_shape_manual(values = c("Burned" = 5, "Unburned" = 0))
```



i. Write a caption

Figure 1. Mean tussock diameter differs across sites. The green open diamonds represent tussock diameters measured (cm) at burned sites, and the pink open squares represent the tussock diameter measured (cm) at unburned sites. The red points represent the means for each site, and lines above and below represent the standard error bars. Data Source: Rocha, A. 2021. Tussock height and diameter in moist acidic tussock tundra at the site of the 2007 Anaktuvuk River fire scar, and nearby unburned

tundra measured in 2016 ver 1. Environmental Data Initiative. https://doi.org/10.6073/pasta/1dccd3fdb3aa693f9c2b69a24f8306ed (Accessed 2026-02-10).

Problem 4. Cleaning, wrangling, and visualization

a. Initial cleaning, wrangling, and summarizing

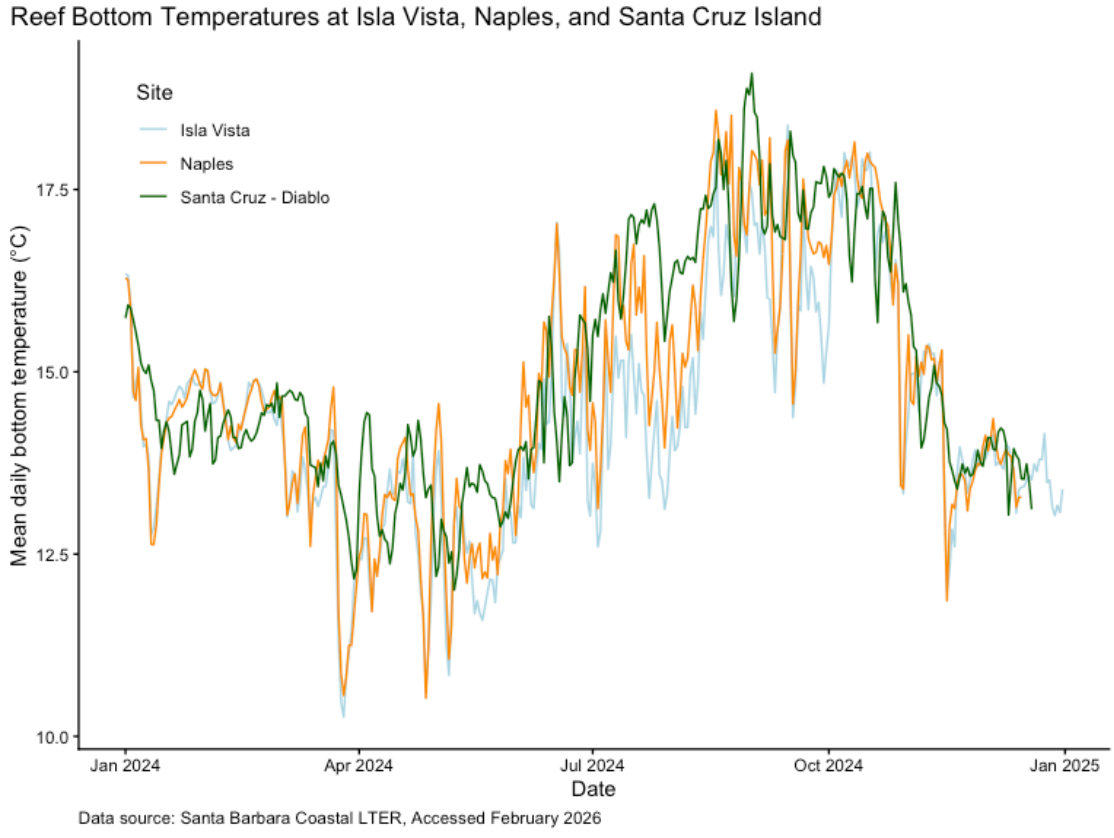
```
temp_clean <- temp |> # start with temp data

 # 1. what package is this function from?
 # the clean_names() function is coming from the janitor package
 clean_names() |>

 # 2. what changes after this function and how?
 # after the clean_names() function, all the column names become lowercase and spaces are
replaced with underscores for clarity



 # 3. what package is year() from?
 # year() comes from the lubridate package in the tidyverse
 # 4. what is this combination of mutate() and year() doing?
 # this creates a new column called "year" that uses only the year value from the "date_local"
column
 mutate(year = year(date_local)) |>

 # 5. what package is filter() from?
 # the filter() function is from the dplyr package in the tidyverse
 # 6. what is being filtered in this functional call?
 # rows are being filtered to keep only observations from three specific sites (NAPL, SCDI,
IVEE) and only from the year 2024
 # 7. what sites and year(s) do you expect to be included in the resulting data frame?
 # only Naples, Santa Cruz-Diablo, and Isla Vista sites from 2024 will be included
filter(site %in% c("NAPL", "SCDI", "IVEE") &
       year == "2024") |>

  # 8. what column is manipulated?
  # the site column
  # 9. what changes about the values in this column?
  # the site acronyms are changed to full names
mutate(site = case_when(
```

```
  site == "IVEE" ~ "Isla Vista",
  site == "NAPL" ~ "Naples",
  site == "SCDI" ~ "Santa Cruz - Diablo"
 )) |>

 # 10. what is being calculated? and at what scale (e.g. annual, monthly, daily)?
 # the mean temperature in °C is being calculated at the daily scale for each site
 # 11. give an example
 # for example, the mean daily temperature in Isla Vista on 2024-01-04 was 14.64829 °C which
is calculated by computing the average temperature in degrees C° (temp_c).
 group_by(site, date_local) |> # group by site
 summarize(mean_daily = mean(temp_c)) |> # calculates the mean
 ungroup() # ungrouping data frame

temp_clean |>
 slice_sample(n = 5) # displays 5 random rows from the "temp_clean" data-frame

# A tibble: 5 × 3
 site              date_local mean_daily
 <chr>             <date>         <dbl>
1 Isla Vista        2024-10-02      16.7
2 Isla Vista        2024-05-10      12.6
3 Isla Vista        2024-05-28      13.5
4 Santa Cruz - Diablo 2024-11-07      14.1
5 Isla Vista        2024-07-31      13.9
```

b. Make the figure

```
# create a plot with cleaned data set
temp_plot <- ggplot(data = temp_clean,
 aes(x = date_local, # x-axis: dates
    y = mean_daily, # y axis mean daily bottom temperatures
    color = site)) +
  # line geometry to show temperature over time
 geom_line() +
 # coloring each site a new color
 scale_color_manual(values = c(
              "Isla Vista" = "lightblue",
              "Naples" = "darkorange",
              "Santa Cruz - Diablo" = "darkgreen")) +
 # label title, x-axis, and y-axis
 labs(title = "Reef Bottom Temperatures at Isla Vista, Naples, and Santa Cruz Island",
```

```
    x = "Date",
    y = "Mean daily bottom temperature (°C)",
    color = "Site",
    caption = "Data source: Santa Barbara Coastal LTER, Accessed February 2026") +
 # provides blank panel background with axis lines
 theme_classic() +
 theme(
  plot.title = element_text(hjust = 0), # left-align title
   plot.title.position = "plot", # Make sure plot title is set all the way to the left
   legend.position = c(0.15, 0.85), # position legend in top left
  legend.background = element_blank(), # remove legend background
  panel.background = element_blank(), # blank panel background
  plot.caption = element_text(hjust = 0) # left-align caption
 )
# displaying plot
temp_plot
```



c. Interpretation

The highest mean daily temperatures are approximately recorded between August and September 2024, visible as the peaks in all three colored lines. The lowest mean daily temperatures recorded are from April to May 2024, shown as the valleys where all three lines dip to their lowest points. The line plot clearly shows this seasonal temperature pattern, with the three colored lines tracking similar seasonal trends across all sites.